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HW:- 1) Find out the applications of recursion.

Is recursion still used in the present programming

If so where?

2) What are the programming languages that support ~~(programming)~~ recursive calls.

1) Recursion is still widely used in present programming. Recursion means, a function calls itself to perform the given task.

Applications :-

① Tree Data structures [Binary Trees, AVL trees]

② Graph Algorithms [DFS, Topological sorting]

③ Divide and Conquer [Merge sort, Quick sort,

④ Backtracking problems - [Sudoku solving, Binary search, Crossword puzzle].

⑤ Dynamic programming (Fibonacci)

⑥ File System operations

⑦ Artificial Intelligence

⑧ Mathematical Computations [Factorial, Tower of Hanoi]

⑨ Fractals and graphics [computer graphics also use recursion].

Yes, it is still used in modern programming era.

(i) OS - Directory traversal, file search

(ii) Compilers - Abstract Syntax Trees (AST's)

- (iii) Databases — for hierarchical data
- (iv) Web browsers — parsing HTML
- (v) Also used in Machine Learning
- (vi) Computer graphics, cloud and DevOps,
- (vii) game development and Networking.

2) Almost all programming languages support recursive calls.

(i) Procedural programming languages

- C
- Pascal

(ii) Object-oriented programming

- Java
- C++
- python
- C#
- Ruby

(iii) Functional programming languages

- Haskell
- Lisp
- Scheme
- F#

(iv) Scripting languages

- Javascript
- Ruby
- python.